Metrology and Robustness

The vocabulary of measurement, and agreement as evidence

Measuring AI Introspection

Overview

Part 1: One measurement, term by term

1. Metrology in one picture
2. Measurand
3. Measurand and realized quantity
4. Indication, model and result
5. Calibration and traceability
6. Uncertainty and its floor
7. Sources of uncertainty
8. Comparability and compatibility
9. Validation for an intended use

Part 2: The introspection experiment

10. Concept injection
11. The experiment in the vocabulary
12. Two senses of calibration
13. Uncertainty in an injection experiment
14. Where the analogy stops: metrology

Part 3: Robustness and convergence

15. Robustness analysis
16. Illusory robustness
17. Independence of methods
18. Perrin's convergence and Hudson's reading
19. Coherence and islands

Part 4: Robustness in AI introspection

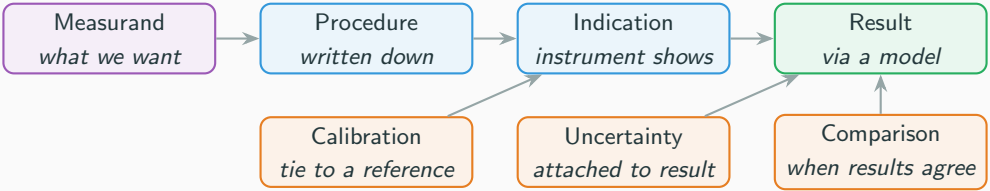
20. A report and a probe on the same activations
21. Convergence, calibration and islands
22. Where the analogy stops: robustness
23. Summary
24. Reading

Part 1

One measurement, term by term

Metrology in one picture

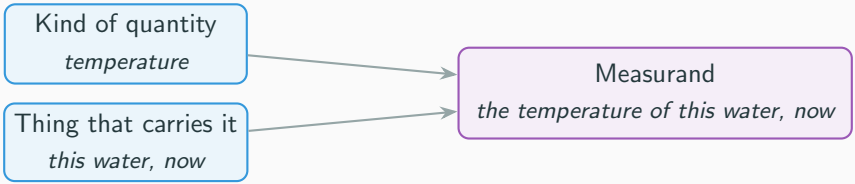
Metrology is the science of measurement. The International Vocabulary of Metrology (VIM) defines its terms, and the Guide to the Expression of Uncertainty in Measurement (GUM) sets the rules for uncertainty.



The measurement card on this site takes its stages from these texts. The GUM adds that a result is complete only with a statement of its uncertainty (3.1.2).

Measurand

The **measurand** is the quantity you intend to measure (VIM 2.3). You specify it by description: the kind of quantity, and the state of the thing that carries it.



How much detail is enough? The GUM's answer (3.1.3): enough that, for the purposes at hand, the measurand has a **unique value**.

A thermometer is the running example for the rest of this part.

Measurand and realized quantity

Three gaps can open between the measurand and what actually gets measured:

Approximation
The measurand often cannot be realized, so you measure a quantity that approximates it.

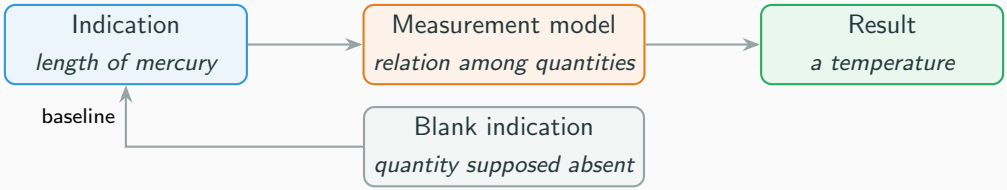
Disturbance
A voltmeter draws current from the battery it measures, so a correction is needed.

Underspecification
Labs measuring "the same" quantity disagree, and it looks like error.

Sources: GUM D.2.1, VIM 2.3 (Note 3), GUM D.3.4. The third gap returns in Part 2: a vague measurand produces disagreement between labs that looks like instrument error.

Indication, model and result

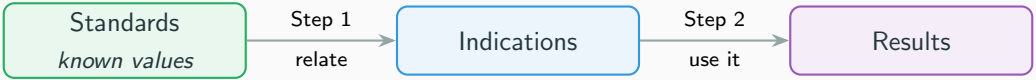
The **procedure** describes the measurement in detail (VIM 2.6). Running it gives an **indication**: the value the instrument provides (VIM 4.1).



The indication belongs to the instrument; the result is a claim about the measurand (VIM 2.9), and the two need not be the same kind of quantity. The **blank indication** (VIM 4.2) is the reading where the quantity is supposed to be absent. It finds the instrument's baseline.

Calibration and traceability

Calibration has two steps (VIM 2.39):



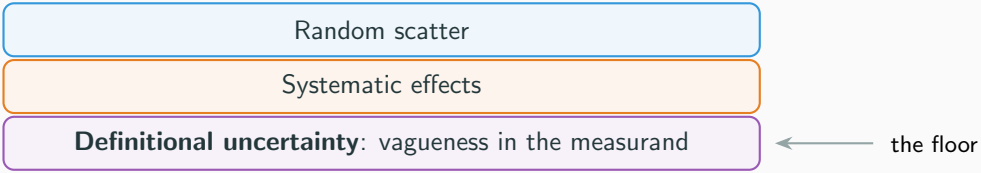
For the thermometer: read it against standards of known temperature, then use that relation to turn a length into a temperature. Step 1 alone is often mistaken for calibration, and so is adjusting the instrument.

Traceability (VIM 2.41) is an unbroken, documented chain of calibrations back to a reference. It does not rule out mistakes, or an uncertainty too large for the purpose.



Uncertainty and its floor

Measurement uncertainty (VIM 2.26) describes the spread of values that could be attributed to the measurand. It covers more than random scatter:



Definitional uncertainty (VIM 2.27) is "the practical minimum" uncertainty achievable for a given measurand.

No instrument can be more precise than the measurand is defined. A better thermometer cannot fix a vague "temperature of this water".

Sources of uncertainty

The GUM lists ten sources (3.3.2). Five of them:

(a)	Incomplete definition of the measurand
(c)	A sample that does not represent it
(d)	Incomplete knowledge of environmental effects
(i)	Approximations in the method
(j)	Variation in repeated observations

Repeating a measurement samples only (j). The GUM asks that all relevant quantities be varied as far as practicable (3.4.2).

Repeatability

Same procedure, operators, system and place, over a short period (VIM 2.20)

Reproducibility

Different places, operators and measuring systems (VIM 2.24)

Comparability and compatibility

Two words that sound alike and ask different questions:

Comparable (VIM 2.46)
Both results are traceable to the same reference. Can they be put side by side?

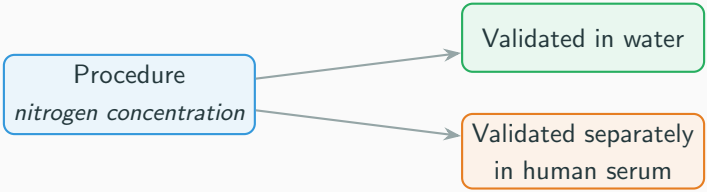
Compatible (VIM 2.47)
They agree within a chosen multiple of the uncertainty of their difference. Do they agree?

The VIM uses compatibility to decide whether two results refer to **the same measurand** (2.47, Note 1).

Two thermometers traced to the same reference are comparable. Whether their readings are compatible is a separate question.

Validation for an intended use

Validation (VIM 2.45) checks that specified requirements are adequate for an intended use. It is always relative to a use.



The VIM's example: a procedure used for nitrogen in water may *also* be validated for nitrogen in human serum. Validation in one setting does not carry over to another automatically.

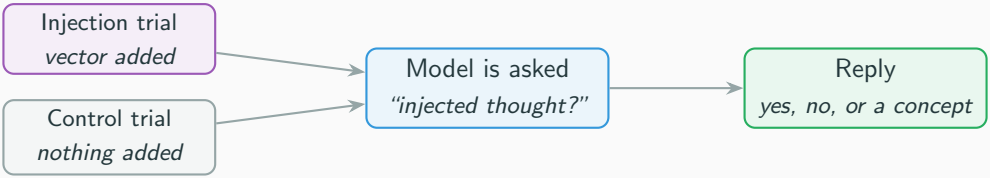
Part 2

The introspection experiment

Concept injection

Lindsey (2025) asks whether a language model notices a change to its own activations:

1. Take a vector for a word such as *bread*.
2. Add it to the model's activations.
3. Ask the model whether it detects an injected thought.



The next slides put this experiment through the vocabulary of Part 1.

The experiment in the vocabulary

Term	Thermometer	Concept injection
Measurand	Temperature of the water	"Introspective access", which admits several readings
Realized quantity	What is actually measured	The injected state, an approximation of the states the claim is about
Indication	A length of mercury	The reply: a string, a yes or no, a number, a choice
Result	A temperature	The claim that the model has access to a state
Blank indication	Quantity supposed absent	Control trials without injection

Lindsey reports no false positives in 100 control trials. The vagueness of "introspective access" is definitional uncertainty, and under the VIM it belongs inside the error bar.

Two senses of calibration

The introspection literature often uses “calibration” in a different sense from the VIM.

Forecasting sense

Do stated probabilities match how often things happen? As in Kadavath et al. (2022).

VIM sense

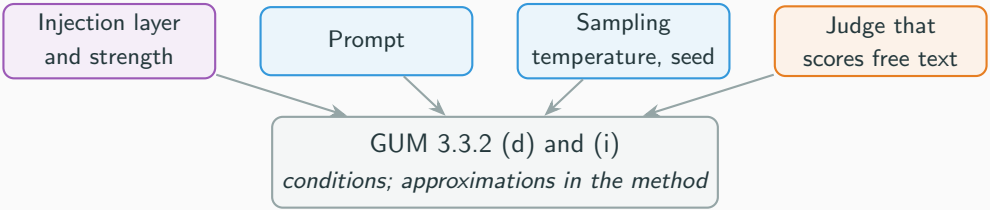
A two-step relation: standards to indications, then indications to results.

The measurement card uses the VIM sense. A model can be calibrated in the first sense without its reports being tied to any reference in the second.

The signal detection page separates the senses further.

Uncertainty in an injection experiment

The influence quantities of an injection measurement:



The GUM asks that these be varied (3.4.2). Resampling at one prompt and one judge samples only source (j).

Changing the judge or the prompt is a change of measuring system: a **reproducibility** condition (VIM 2.24).

Where the analogy stops: metrology

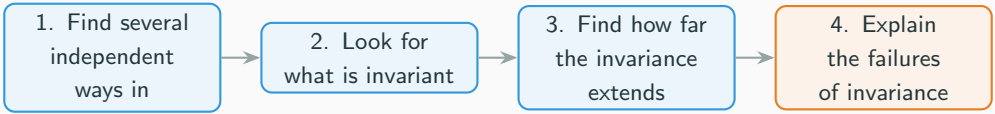
- **Nominal outcomes.** Measurement does not apply to nominal properties (VIM 2.1, Note 1). Many introspection outcomes are nominal, such as which concept was injected, so the vocabulary applies to them only by analogy.
- **No calibration hierarchy.** No reference procedure, standard or unit anchors a report of a model's internal state. Traceability has no chain to run along, and comparability in the VIM sense is unavailable. The card asks what a paper used in place of a reference, and whether the model under test produced it.
- **No model of the disturbance.** The voltmeter correction assumes a model of the disturbance. In self-report the instrument and the object are the same network, and no such model is available.

Part 3

Robustness and convergence

Robustness analysis

Agreement among independent procedures is evidence that a result is real, not an artifact of one procedure. Wimsatt (1981) gives four steps:

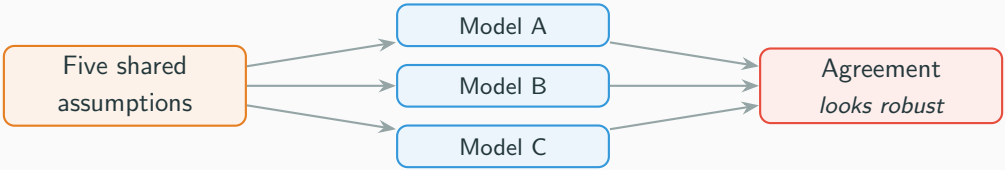


Step 4 treats failures of invariance as information. Wimsatt also says why one route is weak: a quantity reached only one way cannot be tested independently.

Evidence for introspection in a language model often takes this form: a self-report matches a probe, an intervention or a second prompt format.

Illusory robustness

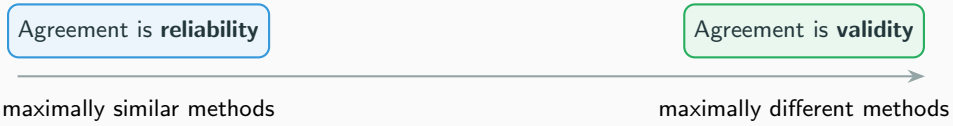
When procedures are presumed independent and are not, the result is **illusory robustness**. Wimsatt calls it the most striking way robustness analysis fails.



His case: group-selection models that agreed because they shared five assumptions, a form of method bias. Such failures are hard to detect. Tal (SEP §8.3) states the measurement version: inaccurate measurements can share a bias, so their errors do not cancel when averaged.

Independence of methods

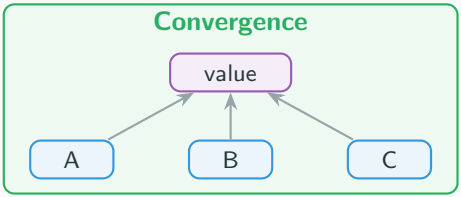
Campbell and Fiske (1959), Wimsatt's source for method bias, treat independence as a matter of degree:



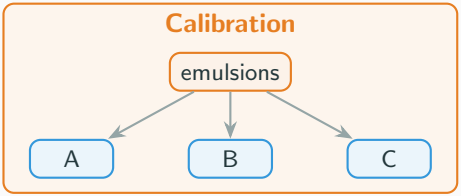
Shared method shows up only when **several traits** are measured by **several methods**; otherwise a high correlation could reflect the trait or the method. The check: a measure should agree more with a different method on the same trait than with the same method on a different trait. The psychometrics page covers their matrix.

Perrin's convergence and Hudson's reading

Jean Perrin determined Avogadro's number from more than a dozen phenomena, such as Brownian motion and radioactivity, and argued that their agreement made molecules nearly certain. Hudson (2013) reads the reasoning differently.



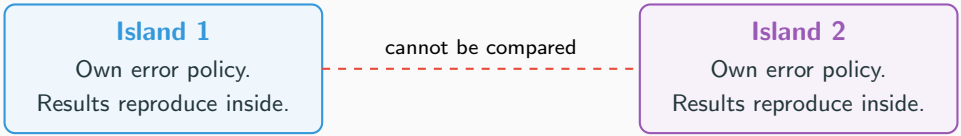
Methods agree as equals, and the agreement supports all of them at once.



Hudson's reading: Perrin's emulsion experiments were the standard. The rest are supported only as far as it is trusted.

Coherence and islands

Tal (2017) treats calibration as a test of invariance across instruments and circumstances. Agreement is not enough: each uncertainty must come from an adequate model of the process.

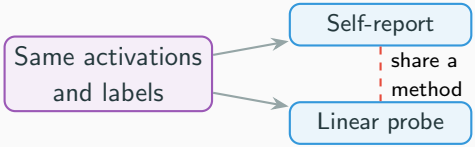


The same indications can be made coherent in more than one way, so a network of instruments can split into islands. Tal's example: the 1990 conventional values for electrical standards, which raised internal consistency and created an island. The remedy is standards that impose one error policy on the network.

Part 4

Robustness in AI introspection

A report and a probe on the same activations



A report and a probe trained on the same activations and labels share part of their method, as do two prompts scored by one judge. Their agreement is exposed to illusory robustness.

Campbell and Fiske's test: a report and a probe on one state should agree more than readings of two different states taken with the same prompt, layer or judge.

	Same state	**Different states**
Different procedures	should agree **more**...	
Same prompt, layer or judge		...than these

Convergence, calibration and islands

Following Hudson, an introspection paper can have either of Perrin's structures:

Convergence
Several procedures agree and none is privileged. The card records it under **comparability**.

Calibration
One procedure, such as injection with a known vector, is the standard. The card records it under **calibration**.

Islands. When each group scores introspection with its own prompts, judges and thresholds, results can reproduce within each setup and fail to compare across setups.

Where the analogy stops: robustness

- **Same substrate.** Perrin's methods rested on different physical phenomena with separate theories. Every procedure for reading a language model's internal state acts on the same network, often at the same layers. Independence has to be argued case by case.
- **No standards body.** Tal's remedy for islands is institutional. No body or standard exists for the internal states of models.

Summary

Idea	Question for an introspection experiment	Card stage
Definitional uncertainty	Which reading of "introspective access"?	Measurand
Blank indication	What do the control trials show?	Instrument
VIM calibration	What stands in for a reference? Did the model make it?	Calibration
Sources of uncertainty	Were layer, prompt, seed and judge varied?	Uncertainty
Compatibility	Do results agree within their uncertainty?	Comparability
Validation	Was the procedure validated for this use?	Validity conditions
Illusory robustness	What do the agreeing procedures share?	Comparability
Hudson on Perrin	Convergence among equals, or one standard?	Calibration
Islands	Do results compare across setups?	Comparability

Reading

- International Vocabulary of Metrology (VIM), JCGM 200:2012: the defined terms.
- Guide to the Expression of Uncertainty in Measurement (GUM), JCGM 100:2008.
- Wimsatt, Robustness, reliability, and overdetermination (1981): illusory robustness.
- Campbell and Fiske, Convergent and discriminant validation (1959): independence of methods.
- Hudson, Seeing Things (2013), ch. 4: Perrin's reasoning as calibration.
- Tal, Calibration: Modelling the measurement process (2017): coherence and islands.
- Tal, Measurement in Science (SEP), §8.3: common bias.
- Lindsey, Emergent introspective awareness in large language models (2025).
- Kadavath et al., Language models (mostly) know what they know (2022).